DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
MONDAY TEST

CLASS : X | **FULL MARKS: 40**
SUBJECT: MATHEMATICS | **DATE: 30.10.24**

General Instructions:

- **The paper consists of three printed pages.**
- **All questions are compulsory.**
- **Copy the question number carefully before answering the questions.**

SECTION: A

1. **[1× 8 = 8]**

i) Given a = $3sec^2\theta$ and b= $3tan^2\theta - 2$. The value of (a – b) is:
a) 1 b) 2 c) 3 d) 5

ii) Which of the following lines cut the positive x-axis and positive y-axis at equal distances from the origin?
a) 3x + 3y=6 b) 5x + 10y = 10 c) -x + y = 1 d) 10x + 5y = 5

iii) If the probability of hitting the target is 0.3, then the probability of not hitting the target is:
a) 0.3 b) 0.5 c) 0.7 d) 1

iv) At a certain time of the day, the ratio of the height of the pole to the length of its shadow is 1: $\sqrt{3}$, then the angle of the sun at that time of the day is:
a) 30^0 b) 45^0 c) 60^0 d) 90^0

v) Assertion (A) : A die is thrown once and the probability of getting an odd number is $\frac{7}{8}$.
Reason (R) : The sample space for odd numbers on a die is {1, 3, 5}

(a) Both A and R are true, and R is the correct reason for A.
(b) Both A and R are true, and R is incorrect reason for A.
(c) A is true, R is false.
(d) A is false, R is true.

vi) If the centre of the circle is (3,5) and the end points of the diameter are (4,7) and (2,y) then the value of y is:
a) -3 b) 3 c) 7 d) 4

vii) Statement 1: $sin^2A + cos^2A = 1$ Statement 2: $sec^2A + tan^2A = 1$
Which of the following is valid?
a) only 1 b) only 2 c) both 1 and 2 d) neither 1 nor 2

viii) Points A(x,y), B(3,-2) and C(4,-5) are collinear. The value of y in terms of x is:

a) 3x -11   b) 11- 3x   c) 3x -7   d) 7 - 3x

SECTION: B

2.

(a) Factorize: $sin^3\theta + cos^3\theta$, hence prove the following identity:

$$\frac{sin^3\theta + cos^3\theta}{sin\theta + cos\theta} + sin\theta\, cos\theta = 1$$

(b) Find:

(i) $(\sin\theta + \operatorname{cosec}\theta)^2$ (ii) $(\cos\theta + \sec\theta)^2$

Using the above results prove the following trigonometry identity.

$(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + tan^2\theta + cot^2\theta$ [4 + 4 = 8]

3.

(a) An integer is chosen between 0 and 100. What is the probability that it is

(i) divisible by 7?

(ii) not divisible by 7?

(iii) a multiple of both 2 and 5?

(b) Each of the letter of the word "ARBITRATION" is written on cards and put in a bag. If a card is drawn at random from the bag after shuffling, what is the probability that the letter on the card is:

(i) a consonant

(ii) one of the letters of the word SPRING.

(iii) not a letter from the word TEAR.

(iv) neither a vowel nor a consonant. [3 +4 =7]

4. Given the equations of two straight lines, L1 and L2 are x – y = 1 and x + y = 5 respectively . If L1 and L2 intersects at point Q (3, 2). Find :

(i) the equation of line L3 which is parallel to L1 and has y-intercept 3.

(ii) the value of k, if the line L3 meets the line L2 at a point P (k,4).

(iii) the coordinate of R and the ratio PQ: QR, if line L2 meets X-axis at point R. [4]

5. Two poles AB and PQ are standing opposite each other on either side of a road 200 m wide. From a point R between them on the road, the angles of elevation of the top of the poles AB and PQ are 45^0 and 40^0 respectively. If height of AB = 80 m, find the height of PQ correct to the nearest metre. [4]

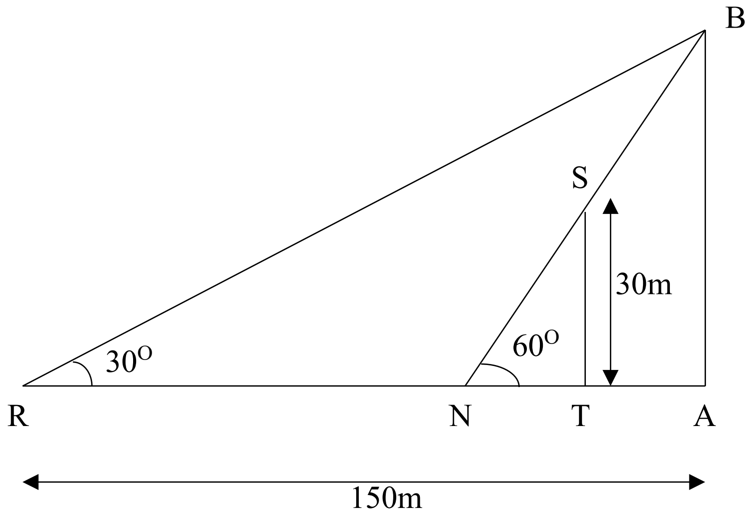
6. A (1, -5), B (2, 2) and C (-2, 4) are the vertices of triangle ABC. Find the equation of :

(i) the median of the triangle through A.

(ii) the altitude of the triangle through B.

(iii) the line through C and parallel to AB. [4]

7.



A tree (TS) of height 30 m stands in front of a tall building (AB). Two friends Rohan and Nisha are standing at R and N respectively, along the same straight line joining the tree and the building (as shown in the diagram). Rohan, standing at a distance of 150 m from the foot of the building, observes the angle of elevation of the top of the building as 30^0. Nisha from her position observes that the top of the building and the tree has the same elevation of 60^0.
Find the:
(a) height of the building
(b) distance between:
(i) Nisha and the foot of the building
(ii) Rohan and Nisha
(iii) Nisha and the tree
(iv) building and the tree. **[5]**